

# Delicate Courier – Financial Operating System (How It Works)

This system is the central financial and operational brain of Delicate Courier. It connects bookings, wallets, deliveries, driver payouts, fuel costs, and external systems (like ShipLogic) into one automated, always-updated system.

## 1. Core Idea

Every real-world action (top-up, booking, delivery, cancellation) becomes a financial event. The system ensures that money is never assumed or manually calculated — it is always recorded, verified, and traceable.

## 2. Wallet System (Client Money Control)

Each client has a digital wallet.

- They deposit money (top-up) via payment provider
- Wallet is only credited AFTER payment confirmation webhook
- Every booking checks wallet balance BEFORE approval
- If insufficient funds → booking is rejected automatically

This prevents debt, unpaid deliveries, and revenue leakage.

## 3. Booking Flow (How a delivery is created)

When a booking request is made:

1. System checks wallet balance
2. System locks funds for the delivery price
3. Booking is either approved or rejected
4. If approved → event is recorded and sent to financial engine + ShipLogic

No booking can exist without financial validation.

## 4. Delivery Completion (Where money is calculated)

When a driver completes a delivery:

- System records actual distance travelled
- Calculates driver earnings
- Calculates fuel cost
- Calculates company profit (margin)
- Updates wallet + ledger automatically
- Sends financial event to external systems

This ensures every delivery immediately reflects real profit and cost.

## 5. Event System (Why nothing is ever lost)

Every action is stored as an “event” in a database queue (Outbox).

- If external systems fail, events are retried automatically
- No data is ever deleted
- Every transaction is replayable and auditable

This guarantees reliability even during outages or API failures.

## 6. Financial Engine Integration

All events are sent to a central financial engine which:

- Maintains full revenue and expense records
- Tracks driver payouts and fuel costs
- Calculates profitability per delivery, driver, and client

This becomes the single source of truth for company finances.

## 7. ShipLogic Integration

Every shipment is also sent to ShipLogic:

- Creates waybills automatically
- Syncs shipment status
- Ensures billing matches real deliveries

This prevents mismatched invoices or missing shipments.

## 8. Profit Visibility (What you can now see instantly)

- Profit per delivery
- Profit per driver
- Profit per bakery/client
- Fuel cost per route
- Real-time company margin

No spreadsheets required.

#### **9. Reliability Layer (Why it is enterprise-grade)**

- Every event is stored before being sent
- System retries failed operations automatically
- Duplicate events are ignored safely (idempotent design)
- Nothing can be double-charged or double-paid

#### **10. What this unlocks**

This transforms Delicate Courier into a fully automated financial operating system where:

- Every delivery becomes a tracked financial transaction
- Cash flow is always accurate and real-time
- Operations and finance are unified
- Scaling does not increase accounting complexity

#### **Final Outcome**

Instead of manually reconciling money and operations, the system continuously updates itself. You always know exactly how much you earned, spent, and profitably delivered — per second, per driver, per client.